METHODS AND VALIDATION

multipgs

Theory, implemented methods, and test results

Package	`multipgs` version 0.3.3
Report date	13 August 2026
Scope	Selection-index theory, three combination routes, deployment, evaluation, automated tests, and committed benchmark results
Evidence base	Unit and contract tests, replicated synthetic experiments, real-LD simulations, and descriptive real-data diagnostics
Automated result	400 tests passed on Python 3.10.20, NumPy 1.26.4, scikit-learn 1.7.2, pytest 9.1.1, and ldpred3 0.4.7
Intended use	Technical explanation and reproducible validation summary; not clinical validation, a regulatory assessment, or a substitute for independent outcome evaluation

Packaged with the `multipgs` source and wheel distributions.

Abstract

`multipgs` combines multiple polygenic scores through three distinct information routes: individual-level penalised stacking, score-space summary-statistic stacking, and a training-free same-trait meta-score. This report derives their common selection-index foundation, states the data and scale contracts for each route, and summarises the package's automated tests and committed benchmark results. The evidence establishes numerical identities and controlled estimator behaviour. It does not establish clinical utility, ancestry portability, or real-world comparative superiority.

Project at a glance

Purpose. The package learns or constructs one deployable polygenic score from several component scores while keeping variant harmonisation, score scale, tuning, and untouched assessment explicit.

Three information routes. Individual-level stacking uses target phenotypes; summary-statistic stacking uses score-space GWAS moments; the training-free route uses same-trait sample-size or expected-accuracy proxies. The routes solve different information problems.

Evidence boundary. Automated tests and committed simulations validate numerical contracts and behaviour under stated designs. External outcomes, population transport, calibration, and decision utility remain the responsibility of each application.

Table 1: Project components and their roles.

Component	Role	Primary output
Panel construction	Align variants and alleles, convert score scales, and represent component scores consistently.	`ScorePanel` and aligned variant weights
Individual fit	Learn a sparse elastic-net combination from target-cohort scores, phenotype, and optional covariates.	Raw-scale score coefficients, cross-validated diagnostics, and frozen variant weights
Summary fit	Learn score coefficients from GWAS cross-moments and an LD-based score Gram matrix.	Standardised and raw-scale coefficients, tuning log, and frozen variant weights
Training-free fit	Construct a same-trait rule from metadata-derived accuracy proxies, optionally using credible target score correlations.	Normalised component coefficients and a combined variant score
Evaluation	Quantify discrimination or incremental prediction on untouched data.	R^2, incremental R^2, liability-scale R^2, AUC, and uncertainty summaries

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1 Project scope and evidence

The package operates after component PGS weights have been obtained. It does not discover variants, estimate GWAS effects, or provide clinical risk calibration. Its central task is smaller: harmonise component scores, combine them using the information available, freeze a deployable variant-weight vector, and evaluate that score on untouched data.

The validation evidence is organised into four levels:

1. algebraic identities and numerical invariants;
2. unit and contract tests;
3. replicated synthetic or real-LD simulations with checked raw rows;
4. descriptive real-data diagnostics without an untouched outcome truth.

These levels are not pooled. A passing simulation establishes behaviour under that simulation, not biological transport or clinical benefit.

The published Multi-PGS paper trained standardised PGS with covariates and reported fivefold out-of-sample assessment [9]. `multipgs` implements the same broad estimator, but substitutes CMSA, adds a nested fallback gate, supports a package-specific summary-statistic objective, and uses a different incremental-R^2 convention. Those differences are made explicit below.

2 Statistical theory

2.1 Notation and score scale

Raw score S_{ik} is the allele-weighted score for individual i and component k. Penalisation is defined after training-data standardisation:

$$Z_{ik} = \frac{S_{ik} - \mu_k}{\sigma_k}, \qquad \beta_k = \frac{\theta_k}{\sigma_k}, \tag{1}$$

where θ_k is the standardised-score coefficient and β_k is the raw-score coefficient returned for deployment. Confusing these coordinates changes the meaning of every penalty factor.

Table 2: Notation used consistently in this report.

Symbol	Meaning
n, q	number of individuals and number of component scores
S, Z	raw and training-standardised score matrices
θ, β	standardised-score and raw-score combination coefficients, related by Equation (1)
g_k, g_f	standardised additive genetic values for component trait k and focal trait f
R_k	$\mathrm{cor}(Z_k, g_k)$, genetic-value accuracy of score k for its own trait
$r_G(k, \ell)$	target-population genetic correlation between traits k and ℓ
ρ	target-population vector with $\rho_k = \mathrm{cor}(Z_k, g_f)$
C	correlation matrix of the standardised component scores in the target population
N_k, M	effective GWAS sample size and effective number of independent effects or segments
π, p_s	population prevalence and sample case fraction for a binary trait; these symbols avoid reusing the score count q

2.2 Selection-index model

A useful working model writes each standardised component score as

$$Z_k = R_k g_k + \sqrt{1 - R_k^2}\, \varepsilon_k, \tag{2}$$

where $\mathrm{var}(\varepsilon_k) = 1$, every ε_k is orthogonal to every genetic value, and the residual errors may be correlated across scores. Under independent residual errors,

$$\rho_k = R_k r_G(k, f), \qquad C_{k\ell} = \begin{cases} 1, & k = \ell, \\ R_k R_\ell r_G(k, \ell), & k \neq \ell. \end{cases} \tag{3}$$

Discovery sample overlap is one mechanism that can correlate the residual errors; shared pipelines, LD references, model fitting, and other common estimation artifacts can do so as well. Conversely, overlap need not induce a large covariance in every design.

For a weight vector w, the squared genetic-value correlation of the linear index is

$$R^2(w) = \frac{(w^\top \rho)^2}{w^\top C w}. \tag{4}$$

When C is positive definite, Cauchy-Schwarz gives

$$w_\star \propto C^{-1}\rho, \qquad R^2_{\max} = \rho^\top C^{-1}\rho. \tag{5}$$

This is the classical selection-index solution [1, 2]. A pseudoinverse or ridge is needed when the score correlation matrix is singular, but neither can repair a misspecified ρ.

2.3 Closed-form gain from one auxiliary trait

For the focal score and one auxiliary score with genetic correlation r, the exact increment above the focal score is

$$\Delta R^2 = \frac{r^2 R_k^2 (1 - R_f^2)^2}{1 - r^2 R_f^2 R_k^2}. \tag{6}$$

The three main drivers are visible: squared genetic correlation, squared auxiliary accuracy, and the squared information still missing from the focal score. A large auxiliary GWAS cannot compensate for genetic irrelevance, and a well-powered focal score leaves less room to gain.

The Daetwyler reference model relates genetic-value accuracy to GWAS power:

$$R_k^2 = \frac{x_k}{1 + x_k}, \qquad x_k = \frac{N_k h_k^2}{M}. \tag{7}$$

It is an upper-reference model under strong architecture and population assumptions, not a per-score truth oracle [3]. The package function `daetwyler_r2` returns phenotypic R^2, $h_k^2 R_k^2$, so taking its square root gives $h_k R_k$.

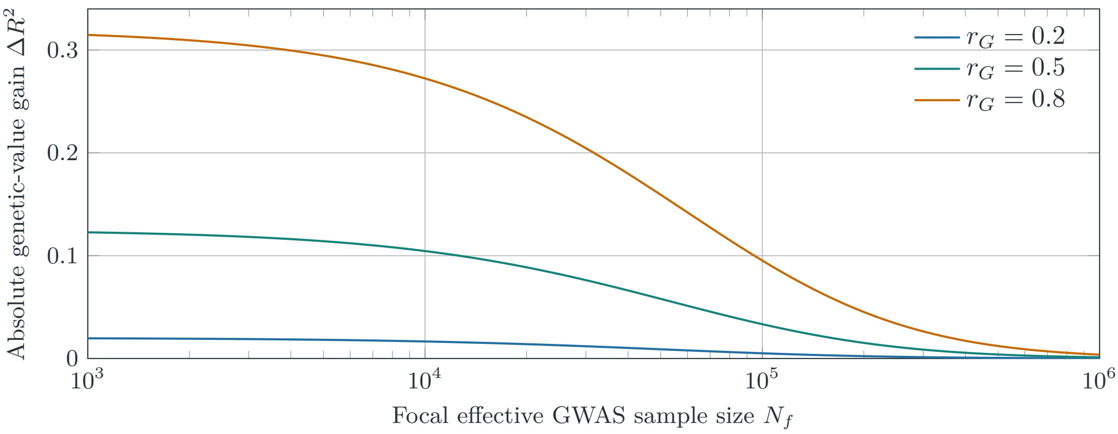

Figure 1: Analytical two-score gain from Equation (6). The illustration fixes $h_f^2 = 0.2$, $M = 20{,}000$, and auxiliary $R_k^2 = 0.5$, so $x_f = N_f/100{,}000$. It is an equation plot, not an empirical performance claim.

3 Implemented methods

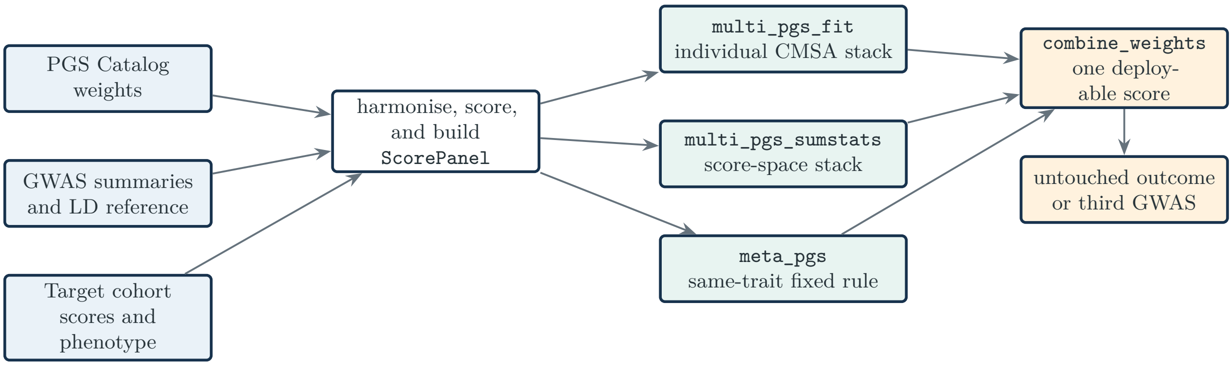

Figure 2: End-to-end package workflow. The three combination routes solve different information problems and are not interchangeable. Assessment data must be untouched by component discovery, combination fitting, and tuning.

Table 3: Information and validation contract for the three fitting routes.

Route	Learns from	Selection / tuning	Valid assessment
`multi_pgs_fit`	Individual score matrix, target phenotype, optional covariates	Inner CMSA choices inside outer folds; final coefficients average fold-selected fits; nested heuristic may return the baseline	Independent individuals who trained neither component scores nor the combination
`multi_pgs_sumstats`	Target-GWAS cross-moments plus external LD score Gram	Same moments, an independent second GWAS, or PUMAS-style pseudo-splits	A third untouched GWAS or independent individual-level cohort; tuning performance is not assessment
`meta_pgs`	Same-trait scores plus effective sample sizes or expected accuracies; target score correlations for decorrelation	No fitted tuning; ridge only stabilises inversion	Independent target outcomes; simple metadata-derived weights remain hypotheses until tested

3.1 Individual-level penalised stacking

Within each training fold, score columns are imputed and standardised. With unpenalised covariates X, write $\eta_i = b_0 + X_i^\top\gamma + Z_i^\top\theta$. The Gaussian or binomial elastic-net objective is

$$\min_{b_0,\gamma,\theta} \underbrace{\begin{cases} (2n)^{-1}\sum_i (y_i-\eta_i)^2, & \text{Gaussian,} \\ -n^{-1}\sum_i\{y_i\eta_i - \log(1+\exp\eta_i)\}, & \text{binomial,}\end{cases}}_{\mathcal{Q}(y,\eta)} + \lambda\sum_{k=1}^{q} p_k\left\{\alpha|\theta_k| + \frac{1-\alpha}{2}\theta_k^2\right\}. \tag{8}$$

Covariates receive penalty factor zero. The Gaussian path uses covariance updates; the binomial path uses IRLS with coordinate descent, following the glmnet family of algorithms [4]. For a standardised Gaussian column with current partial score $z_k^\star$, the coordinate update is

$$\theta_k \leftarrow \frac{\operatorname{sign}(z_k^\star)(|z_k^\star| - \lambda\alpha p_k)_+}{G_{kk} + \lambda(1-\alpha)p_k}. \tag{9}$$

CMSA fits a path inside folds, selects a fold-specific model, and averages the selected coefficient vectors [6]. `multipgs` adds a separate nested outer-fold decision rule that falls back to the unpenalised baseline when the apparent increment is insufficient. The null benchmark shows that this is a pragmatic guard, not a calibrated hypothesis test: pure noise passed in 11 of 60 checked runs overall.

The averaged support is the union of fold supports. Consequently `selected()` reports predictive weights, not causal traits, unique genetic contributors, or stable feature-selection discoveries. In the checked simulation, support recall is high (0.84–1.00 by regime) while precision is only 0.09–0.20.

3.2 Summary-statistic score-space stacking

Let W_{LD} contain component weights on the LD source's standardised-genotype scale, D be its LD correlation matrix, and W_{GWAS} contain the same component scores on the GWAS source's standardised-genotype scale.

The sufficient moments are

$$G_{\text{raw}} = W_{\text{LD}}^{\mathsf{T}} D W_{\text{LD}}, \qquad c_{\text{raw}} = W_{\text{GWAS}}^{\mathsf{T}} z. \tag{10}$$

Writing $A = \text{diag}(\sqrt{\text{diag}(G_{\text{raw}})})$, the fitted coordinates use

$$G = A^{-1} G_{\text{raw}} A^{-1}, \qquad c = A^{-1} c_{\text{raw}}. \tag{11}$$

This is the same Gaussian objective as Equation (8), but in the span of the supplied scores. At $\alpha = 1$, the lasso selects entire component scores; it is inspired by the SNP-level lassosum quadratic [5], but is not SNP-level lassosum.

Equations (10)–(11) are exact sample moments for unadjusted centred data, or when genotypes and phenotype were jointly residualised on the same covariate design. Ordinary covariate-adjusted marginal GWAS coefficients do not alone establish that identity. With external LD, G and c are source-specific plug-in estimates rather than one empirical covariance matrix.

For a fixed coefficient vector, summary tuning evaluates

$$\text{MSE}(\theta) = \text{var}(y) - 2\theta^{\mathsf{T}} c + \theta^{\mathsf{T}} G \theta, \qquad \widetilde{R}^2(\theta) = \frac{(\theta^{\mathsf{T}} c)^2}{\text{var}(y)\, \theta^{\mathsf{T}} G \theta}. \tag{12}$$

MSE, not squared correlation, selects the path because R^2 is blind to a reversed predictor. An independent second GWAS can tune the path; a third is still needed for assessment. PUMAS-style pseudo-splitting uses a joint-Gaussian/CLT covariance approximation [7]. In the committed calibration, relative covariance error is 3.77% for Gaussian and 10.99% for binary simulations. Every scalar and pseudo-split path carries coordinate-descent status; iteration-exhausted candidates are excluded rather than silently selected.

3.3 Training-free same-trait combination

The simple rules standardise component scores and use

$$w_k \propto \sqrt{N_{\text{eff},k}} \quad \text{or} \quad w_k \propto \sqrt{\widehat{R}_k^2}. \tag{13}$$

The first is a power-limited approximation sourced from pipeline code, not from the Hansen et al. preprint. Under independent estimation errors, the exact same-trait GLS direction is

$$w_k \propto \frac{R_k}{1 - R_k^2}, \tag{14}$$

so direct accuracy weighting is accurate only as every R_k approaches zero. The decorrelated rule is

$$w_\delta \propto (C + \delta I)^{-1} \rho. \tag{15}$$

Its algebra is exact under Equation (5), but small error in ρ is amplified along low-eigenvalue directions of C. Ridge reduces that amplification only by moving the result back toward an undecorrelated weighted sum. If only Daetwyler proxies are available, the checked real CAD diagnostic supports `expected_r2`, not `decorrelated`.

3.4 Panel construction, deployment, and evaluation

`panel_from_catalog` harmonises alleles and constructs scores in one genotype pass. `panel_from_sumstats` fits component scores through ldpred3. Identifier order, genome build, effect orientation, and source-specific dosage standard deviation are part of the statistical contract, not file-format trivia.

Write component k's term as $c_{jk}(g_j - \mu_{jk})$, with $c_{jk} = w_{jk}/\sigma_{jk}$ for a standardized score and $c_{jk} = w_{jk}$ for a raw score. Define $C_j = \sum_k \beta_k c_{jk}$ and $M_j = \sum_k \beta_k c_{jk} \mu_{jk}$. The exact one-row frozen representation has

$$\mu_j^\star = \frac{M_j}{C_j}, \qquad \widetilde{w}_j = \sigma_j^\star C_j. \tag{16}$$

Raw allele-count components use the target mean and leave one removable intercept. If $C_j = 0, M_j \neq 0$, or $\mu_j^\star \notin [0, 2]$, no valid single LDpred3 row can preserve both observed dosage and reference-mean missing imputation, so deployment is refused rather than silently rescaled. The frozen scoring scale should be carried to new cohorts. Re-standardising each cohort independently can change coefficients, calibration, and even the meaning of a fixed threshold.

With covariates, `multipgs` reports the unnormalised increment

$$R^2_{\text{incremental}} = R^2_{\text{covariates+score}} - R^2_{\text{covariates}}. \tag{17}$$

The Albiñana paper instead divides this difference by $1-R^2_{\text{covariates}}$ [9]; absolute values are therefore not identical even when relative comparisons share covariates.

For binary outcomes, with population prevalence π, sample case fraction p_s, threshold $t = \Phi^{-1}(1-\pi)$, density $\varphi(t)$, and $i = \varphi(t)/\pi$, the implemented Lee transformation is

$$C_{\text{LT}} = \frac{[\pi(1-\pi)]^2}{\varphi(t)^2 p_s(1-p_s)}, \quad \vartheta = \frac{i(p_s-\pi)}{1-\pi}\left\{\frac{i(p_s-\pi)}{1-\pi} - t\right\}, \quad R^2_L = \frac{C_{\text{LT}} R^2_O}{1 + C_{\text{LT}}\vartheta R^2_O}. \tag{18}$$

This metric depends on a stated prevalence assumption [8].

4 Validation results

4.1 Individual fitting: estimator sanity

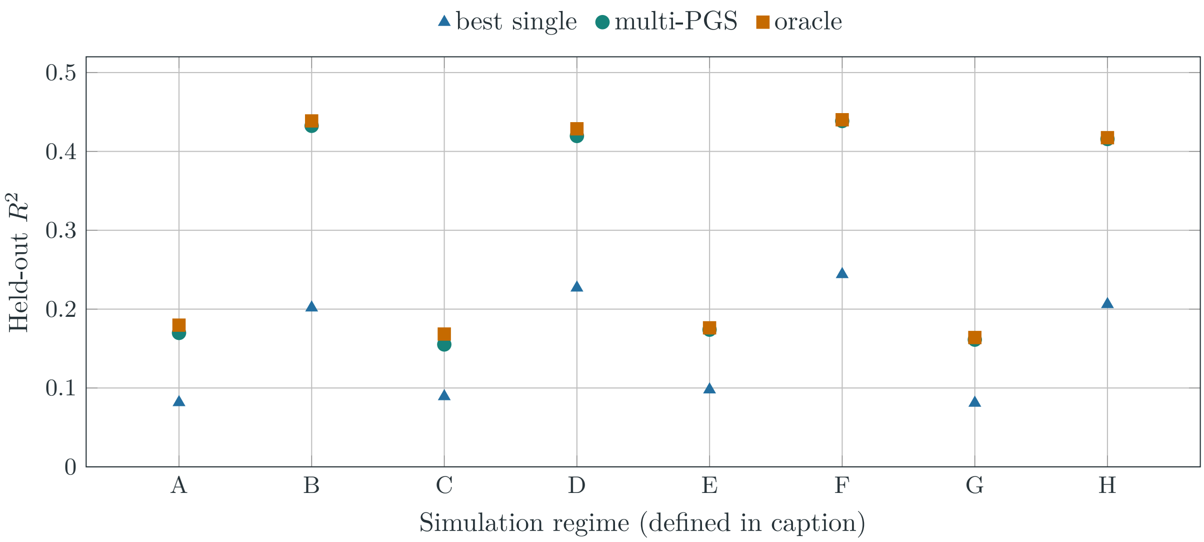

Figure 3: Mean held-out R^2 over 30 seeds from `benchmarks/results/fit_accuracy_summary.csv`. Regimes A–D use $n = 2000$, E–H use $n = 10{,}000$; A,B,E,F use $q = 50$, the others $q = 200$; A,C,E,G use $h^2 = 0.2$, the others $h^2 = 0.5$. The multi-PGS beats the training-selected best single score in 240 of 240 raw replicates. Because the simulated genetic value lies in the score span, this is an estimator sanity check, not evidence of clinical efficacy.

Across the eight regimes, mean uplift over the selected single score ranges from 0.0660 to 0.2307 R^2. Based on regime means, the fitted stack closes 83.4%– 99.2% of the gap between the selected single score and the oracle. The overall mean nested-CV minus held-out difference is -0.00230, slightly conservative in this simulation.

4.2 Summary moments and LD-reference sensitivity

With exact individual moments, the summary and individual standardised coefficients correlate 0.999858 on average, and all compared held-out R^2 values lie between 0.4973 and 0.4983. This supports the score-space identity under the benchmark's strong-signal, small-q design.

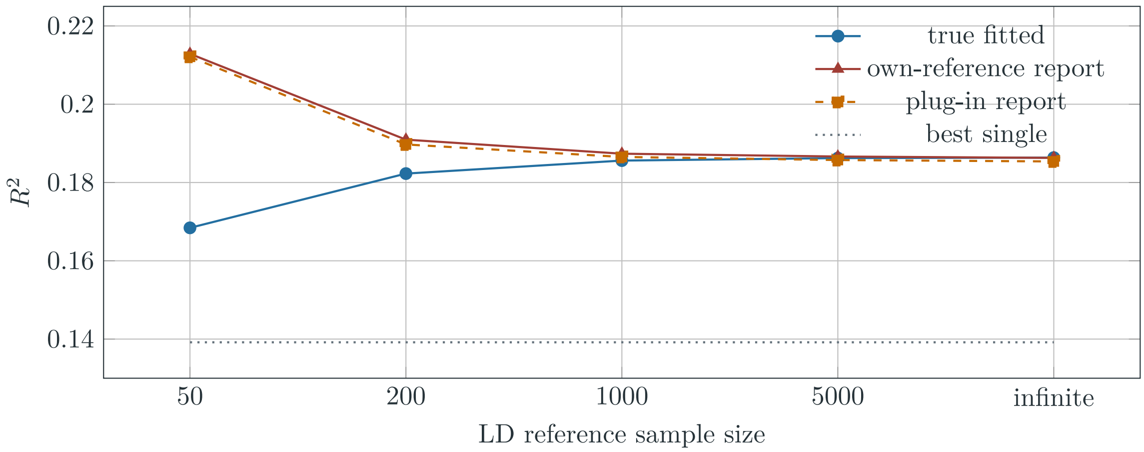

Figure 4: Real-LD simulation from `real_ld_simulation_summary.csv`. At reference $n = 50$, true fitted R^2 falls by 0.01795 relative to the infinite-reference result, while the own-reference estimate is inflated by 0.04446 (26.4% of true fitted R^2). Reporting bias is larger than the loss in the fitted predictor.

4.3 Overlap and decorrelation

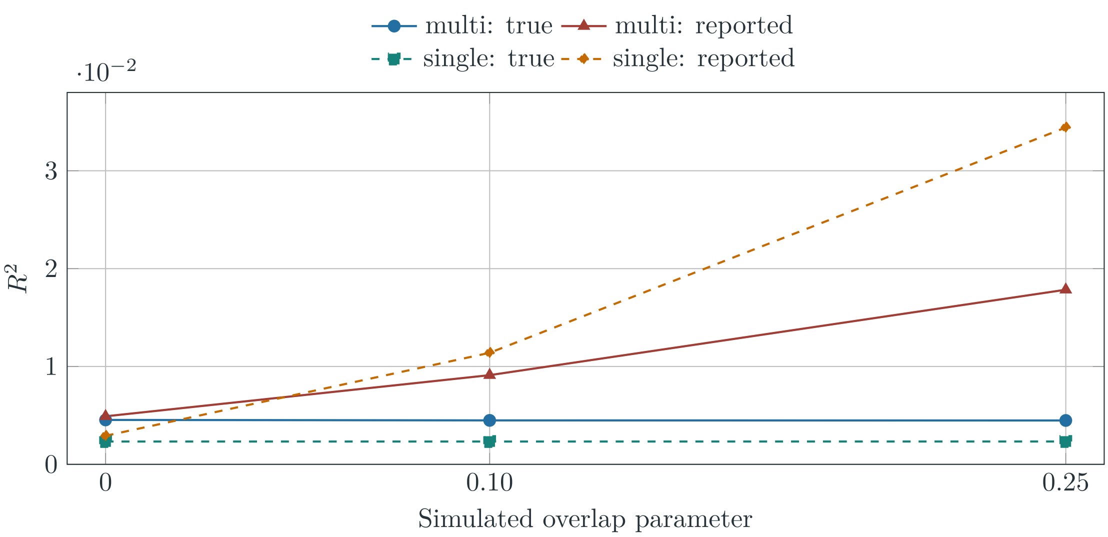

Figure 5: Known-overlap simulation from `overlap_inflation_summary.csv`. At overlap 0.10, reported-to-true R^2 ratios are 2.03 for multi-PGS and 4.89 for the single score; at 0.25, they are 3.98 and 14.73. Bivariate LDSC detected 16 of 16 non-null runs and 0 of 8 null runs in this 109,590-variant design. This does not imply universal LDSC power, but it refutes the absolute claim that no downstream diagnostic can ever flag overlap.

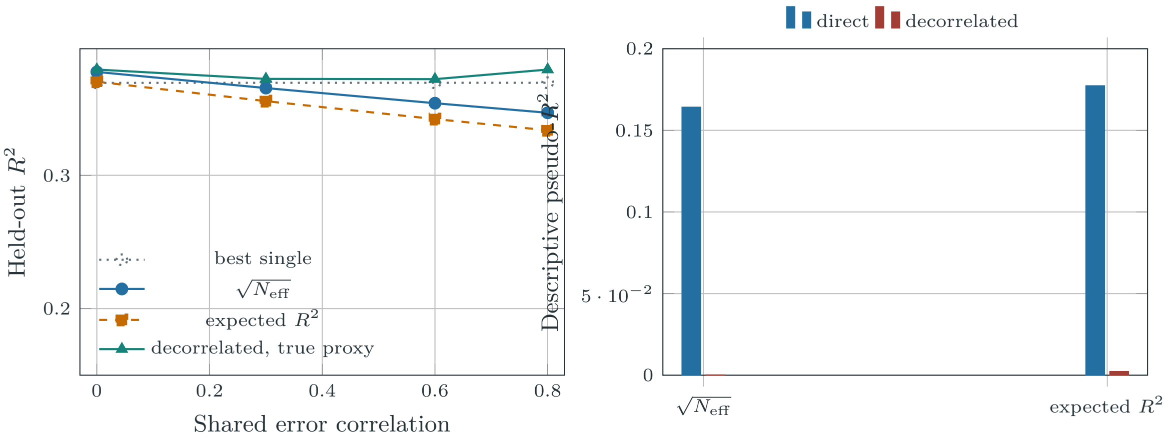

Figure 6: Decorrelation behaves well when the simulation supplies the intended accuracy structure (left, 30 seeds per point), but collapses with proxy accuracies in the single checked 24-score CAD diagnostic (right). The right panel is a contaminated regime-C score-space description, not external outcome truth. Sources: `meta_rules_summary.csv` and `real_meta_rules_summary.csv`. The direct expected-R^2 proxy is 0.17729, versus 0.002261 after decorrelation.

Table 4: Headline validation evidence, separated by evidence level.

Artifact	Result	Interpretation
`fit_accuracy`	Multi-PGS wins 240/240 held-out simulations; uplift 0.0660–0.2307	Supports estimator sanity when truth lies in the score span; does not establish real-outcome or clinical superiority.
`null_gate`	Noise passes 11/60 runs overall	Shows the fallback is a decision heuristic, not a significance test.
`sumstat_vs_individual`	coefficient correlation 0.999858; R^2 0.4973–0.4983	Supports exact-moment Gaussian equivalence in a small-q, strong-signal design.
`sumstat_calibration`	PUMAS covariance relative error 3.77% Gaussian, 10.99% binary	Quantifies approximation error; it does not validate pseudotuning as assessment.
`real_ld_simulation`	reference $n = 50$ report inflation 0.04446	Demonstrates reference-induced optimism under real LD.
`overlap_inflation`	up to 3.98-fold multi-PGS and 14.73-fold single-score inflation	Demonstrates overlap danger under a known covariance construction.
`real_meta_rules`	direct expected-R^2 proxy 0.17729 versus decorrelated 0.002261	Shows severe proxy sensitivity in one regime-C CAD diagnostic; cannot rank methods externally.

5 Automated verification

The complete automated suite passed 400 of 400 tests in the environment stated on the title page. It covers public API validation, allele and variant alignment, score-scale conversion, Gaussian and binomial fitting, summary moment construction, independent and pseudo-tuning, meta-score rules, evaluation metrics, serialization, documentation links, benchmark metadata, and distribution contents. Numerical regressions include full-rank and rank-deficient moment projections, positive score rescaling, exact-moment equivalence, solver iteration status, and invalid-control rejection.

Table 5: Automated verification layers and their interpretation.

Layer	Representative assertions	Result
Algebra and numerics	Standardisation and back-transformation, coordinate updates, Gram ranges, projection identity and idempotence, and score-rescaling invariance	Passed in the full test suite
Fitting contracts	Shape and scale validation, tuning separation, frozen weights, convergence logs, deterministic seeds, and baseline fallback	Passed in the full test suite
Genomic data handling	Allele orientation, ambiguous variants, duplicate handling, LD/GWAS scale agreement, and missing-data behaviour	Passed in the full test suite
Evaluation and documentation	Continuous and binary metrics, liability conversion, cited resources, benchmark result consistency, and example paths	Passed in the full test suite
Distributions	Expected source and wheel contents, installed import, and command-line smoke checks	Passed in the package build checks

These tests establish implementation contracts; the replicated benchmark experiments in Section 4 establish statistical behaviour under their stated data-generating processes. Neither substitutes for application-specific external validation.

6 Interpretation limits

1. **No causal interpretation.** Coefficients optimise prediction under correlation and penalisation. They do not estimate genetic correlations, mediation, or trait-specific causal contributions.
2. **No automatic overlap cure.** Cross-validation inside a target cohort cannot remove discovery contamination shared by every fold. Bivariate LDSC or named-cohort metadata may flag overlap under favourable conditions, but exclusion by design is stronger.
3. **No ancestry model.** Fitting weights in a target cohort does not repair discovery-score or LD-reference non-portability. Absolute risk and score means are particularly unsafe across ancestries.
4. **Direction matters.** Squared correlation hides sign reversal, and the training-free expected-R^2 routes assume consistent positive trait orientation. Report signed correlation or calibration slope alongside R^2.
5. **No clinical calibration.** A high ranking metric is not a calibrated probability. Calibration intercept/slope, subgroup performance, decision utility, and an externally justified prevalence are separate requirements.
6. **Conditional uncertainty.** The package bootstrap conditions on fixed component scores and their discovered weights; it does not propagate component-GWAS, LD-reference, or architecture-estimation uncertainty.

7 Conclusion

`multipgs` has a coherent central idea: move a difficult multi-trait genomic problem into a smaller space of component scores, then choose weights using the information actually available. The selection-index derivation explains when gains are possible; the individual and summary Gaussian objectives explain how weights are learned; and the same-trait rules provide a transparent fallback when phenotypes are unavailable.

The choice among routes is determined by the information available, not by a single universal ranking. Individual fitting is appropriate when a target phenotype and honest resampling are available. Summary fitting preserves the same score-space structure when GWAS cross-moments and LD are available. Training-free rules are transparent metadata-based constructions whose accuracy assumptions must be checked externally. Across all routes, the deployable object is a fixed variant-weight vector and the final scientific claim comes from untouched, population-relevant outcome data.

The committed tests and benchmarks demonstrate scale handling, numerical equivalence, controlled prediction gains, null behaviour, and sensitivity to LD reference quality and discovery overlap. They explain how the software behaves under known conditions. They deliberately stop short of claiming clinical validity, portability, or calibrated risk.

References

[1] H. F. Smith. A discriminant function for plant selection. *Annals of Eugenics*, 7(3):240–250, 1936. doi:10.1111/j.1469-1809.1936.tb02143.x.

[2] L. N. Hazel. The genetic basis for constructing selection indexes. *Genetics*, 28(6):476–490, 1943. doi:10.1093/genetics/28.6.476.

[3] H. D. Daetwyler, B. Villanueva, and J. A. Woolliams. Accuracy of predicting the genetic risk of disease using a genome-wide approach. *PLoS ONE*, 3(10):e3395, 2008. doi:10.1371/journal.pone.0003395.

[4] J. Friedman, T. Hastie, and R. Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 2010. doi:10.18637/jss.v033.i01.

[5] T. S. H. Mak, R. M. Porsch, S. W. Choi, X. Zhou, and P. C. Sham. Polygenic scores via penalized regression on summary statistics. *Genetic Epidemiology*, 41(6):469–480, 2017. doi:10.1002/gepi.22050.

[6] F. Privé, H. Aschard, and M. G. B. Blum. Efficient implementation of penalized regression for genetic risk prediction. *Genetics*, 212(1):65–74, 2019. doi:10.1534/genetics.119.302019.

[7] Z. Zhao, Y. Wu, J. Miao, Y. Song, X. Zhong, et al. PUMAS: fine-tuning polygenic risk scores with GWAS summary statistics. *Genome Biology*, 22:257, 2021. doi:10.1186/s13059-021-02479-9.

[8] S. H. Lee, M. E. Goddard, N. R. Wray, and P. M. Visscher. A better coefficient of determination for genetic profile analysis. *Genetic Epidemiology*, 36(3):214–224, 2012. doi:10.1002/gepi.21614.

[9] C. Albiñana, Z. Zhu, A. J. Schork, et al. Multi-PGS enhances polygenic prediction by combining 937 polygenic scores. *Nature Communications*, 14:4702, 2023. doi:10.1038/s41467-023-40330-w.

[10] O. S. Hansen, L. A. Rasmussen, E. M. Pedersen, et al. Mapping genetic architecture of thousands of complex traits using GWAS summary statistics. *Research Square*, version 1, 2026. Preprint; not peer reviewed. doi:10.21203/rs.3.rs-9415305/v1.

A Artifact provenance

This report was generated in the context of source revision `49d8e71f0006`. Its reproducibility authority is the SHA-256 over the declared report inputs: `7164e45ff7439036f16c0ea0dabd2bb8db5e0112b5e8b11eb9a318acdf53a91e`. The evidence generator records that input list, the test environment, and benchmark-derived headline values; generated files and the PDF are excluded to avoid a self-referential digest.

Table 6: Files used directly for quantitative statements and figures.

Repository artifact	Use in this report
`benchmarks/results/fit_accuracy.csv` and `fit_accuracy_summary.csv`	Individual-fit held-out accuracy, uncertainty, support, and CV calibration.
`benchmarks/results/null_gate.csv` and `null_gate_summary.csv`	Pure-noise and signal fallback behaviour.
`benchmarks/results/sumstat_vs_individual_summary.csv`	Individual/summary coefficient and held-out accuracy agreement.
`benchmarks/results/sumstat_calibration_summary.csv`	Exact-moment identity and PUMAS covariance approximation error.
`benchmarks/results/real_ld_simulation_summary.csv`	Real-LD reference-size sensitivity and reporting bias.
`benchmarks/results/overlap_inflation_summary.csv`	Known-overlap accuracy inflation and LDSC diagnostic rate.
`benchmarks/results/meta_rules_summary.csv`	Synthetic same-trait rule sensitivity over shared residual correlation.
`benchmarks/results/real_meta_rules_summary.csv`	Single real CAD regime-C descriptive comparison.

Three historical real-data provenance records report ldpred3 0.3.1, whereas the package now requires ldpred3 0.4.7–0.4.x. Their quantitative results are reported as historical diagnostics under the recorded environments, not as fresh current-version runs.